CROPGUARD AI- DEEP LEARNING BASED ON PLANT DISEASE DETECTION WITH GUIDANCE.

Abstract

The Plant Disease Detection and Supplement Recommendation System is an intelligent web-based application designed to assist farmers and gardeners in early detection of plant diseases and provide effective treatment solutions. Leveraging a Convolutional Neural Network (CNN) trained on 27 classes of plant diseases, the system accurately identifies diseases from leaf images uploaded by users. The application processes the uploaded image, applies preprocessing and feature extraction through the CNN, and predicts the disease with high accuracy. In addition to disease identification, the system offers detailed information including the disease name, description, preventive measures, and visual references. Furthermore, it provides personalized supplement and fertilizer recommendations along with purchase links, helping users maintain plant health and prevent crop loss. The system is implemented using Flask for a responsive web interface, PIL for image processing, and PyTorch for deep learning model deployment. Supplement and disease data are dynamically loaded from CSV files, ensuring easy scalability and updates. This platform combines deep learning, computer vision, and real-time web technologies to create a practical, user-friendly tool that empowers users to monitor, diagnose, and manage plant health efficiently, ultimately improving agricultural productivity.

Introduction

Agriculture is a vital sector that supports human survival, economic stability, and national growth by providing food and livelihoods to millions worldwide. However, plant diseases caused by fungi, bacteria, viruses, and nutrient deficiencies severely affect crop yield, quality, and market value, leading to financial losses for farmers. With advancements in Artificial Intelligence (AI), Deep Learning (DL), and Computer Vision, automated plant disease detection systems have become effective solutions. Convolutional Neural Networks (CNNs) can analyze leaf images and identify disease patterns such as color changes, spots, and texture variations with high accuracy.

To address these challenges, CropGuardAI is introduced as a web-based platform that uses deep learning for real-time plant disease detection. Farmers can upload images of infected leaves, and the system can identify up to 27 types of diseases quickly and accurately.

Beyond detection, the platform provides detailed guidance, including causes, symptoms, prevention methods, and treatment recommendations. It also suggests suitable fertilizers and supplements to improve crop health.

Existing System

- ❖Most current plant disease detection systems focus on only one disease using basic machine learning algorithms such as K-Nearest Neighbors (KNN).
- ❖They rely heavily on manual feature extraction like leaf texture, color, and shape analysis.
- ❖Users often need expert knowledge to interpret the results accurately.
- ❖These systems do not provide treatment guidance or supplement/fertilizer recommendations.
- ❖They generally have lower accuracy and cannot handle multiple disease classes efficiently.
- ❖Existing solutions are not scalable and cannot be easily updated to include new diseases or treatments.
- ❖They typically lack user-friendly web interfaces, making them less accessible for general users or farmers.

Drawbacks of Existing System:

- ❖Limited to detecting only one disease per system.
- ❖Lower accuracy compared to deep learning-based methods.
- ❖Requires expert knowledge for interpreting results.

Proposed System

The proposed system is a web-based platform that automates plant disease detection and provides actionable supplement recommendations to users. Using a trained Convolutional Neural Network (CNN), the system analyzes images of plant leaves uploaded by farmers or gardeners to accurately identify the type of disease. Once detected, it displays comprehensive information about the disease, including its name, description, preventive measures, and reference images, along with suggestions for suitable supplements or fertilizers and links to purchase them. The system is implemented using Flask for a responsive web interface, PyTorch for model deployment, and Python libraries for image processing and data management. By integrating disease detection with supplement guidance in a single platform, the system aims to assist users in monitoring plant health proactively, reduce crop loss, and enhance agricultural productivity efficiently.

Benefits of Proposed System:

❖ Enables early and accurate detection of plant diseases, reducing crop loss.

❖ Provides detailed information about diseases, including preventive measures and reference images.

❖ Recommends suitable supplements and fertilizers with direct purchase links, simplifying treatment.

❖ User-friendly web interface allows easy image upload and quick access to results.

❖ Reduces dependency on expert intervention, making plant care accessible to all users.

Software Modules

Module 1: User Interface & Image Upload Module

This module provides a responsive and easy-to-use web interface developed using Flask, HTML, CSS, and Bootstrap. It enables users to upload plant leaf images for disease analysis through a browser-based form. The module validates image formats and ensures secure data transfer between the client and server. By offering a simple navigation structure, it allows farmers and gardeners to interact with the system efficiently without requiring technical knowledge.

Module 2: Image Preprocessing Module

The image preprocessing module prepares uploaded leaf images for accurate disease detection. Using the Python Imaging Library (PIL), images are resized to the required input dimensions, normalized, and converted into appropriate formats for model processing. This module enhances image quality and consistency, reduces noise, and ensures uniformity across inputs, which improves the performance and accuracy of the CNN-based disease detection model.

Module 3: Plant Disease Detection Module (CNN Model)

This module is the core of the system and uses a Convolutional Neural Network trained on 39 classes of plant diseases. Implemented using PyTorch, the model automatically extracts meaningful features from leaf images and classifies them into disease categories. The trained CNN performs real-time inference with high accuracy, enabling early disease detection and supporting timely decision-making for effective plant disease management.

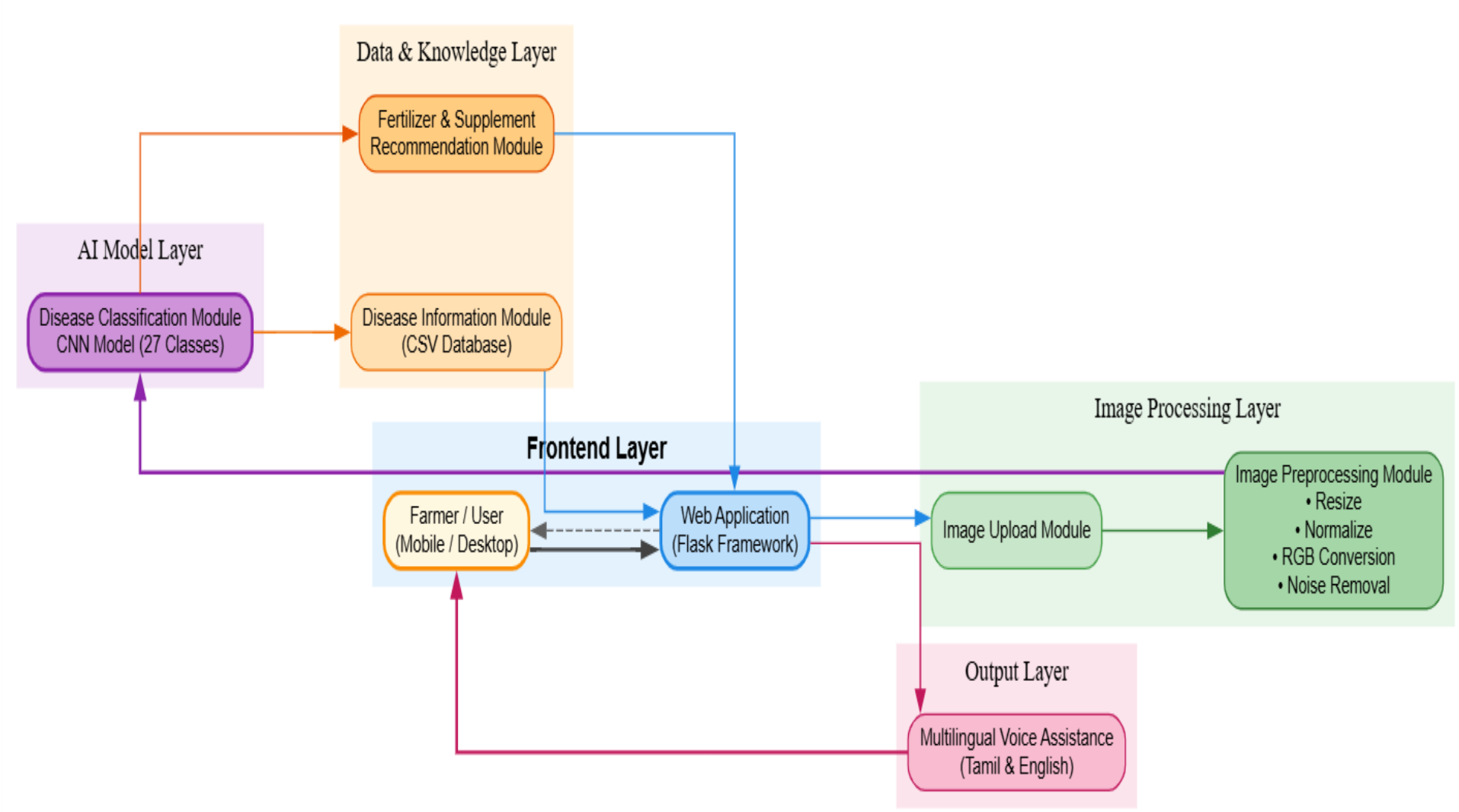
Module 4: Disease Information & Analysis Module

After disease prediction, this module retrieves detailed information related to the identified plant disease from CSV-based datasets. It displays the disease name, description, symptoms, causes, preventive measures, and reference images. This information helps users understand the nature and severity of the disease. By providing clear guidance, the module assists users in taking preventive actions to reduce further spread and damage.

Module 5: Supplement & Fertilizer Recommendation Module

This module provides personalized supplement and fertilizer recommendations based on the detected plant disease. The recommendations are dynamically loaded from CSV files and include treatment details, usage instructions, and purchase links. This module helps users apply appropriate supplements at the correct time, improving plant recovery and minimizing crop loss. It enhances decision-making by combining disease detection with actionable treatment solutions.

Architecture Diagram



Database Design

Field Name	Data Type	Description
user_id	INT	Unique user ID
user_name	VARCHAR	Name of the user
image_id	INT	Unique image ID
image_path	VARCHAR	Path of uploaded leaf image
disease_id	INT	Unique disease ID
disease_name	VARCHAR	Name of detected disease
cause	TEXT	Cause of disease
symptoms	TEXT	Symptoms of disease
prevention	TEXT	Prevention methods
fertilizer_name	VARCHAR	Recommended fertilizer/supplement
usage_details	TEXT	Instructions for usage
confidence_score	FLOAT	Prediction accuracy
audio_path	VARCHAR	Voice output file path
created_at	DATETIME	Date and time of record

System Testing

1. Unit Testing :

Unit testing was carried out on individual components such as image upload, preprocessing, disease prediction model, CSV reading, and supplement recommendation modules. Each function was tested separately to check logical errors, performance, data handling, and exception management. This ensured every module worked correctly before integration into the main system.

2. Integration Testing :

Integration testing focused on verifying smooth interaction between system modules. The Flask interface, PyTorch CNN model, and CSV-based databases were combined and tested. Data flow, communication accuracy, and response handling were checked to ensure that once modules were connected, they worked together correctly without conflicts, crashes, or data loss.

3. Performance Testing :

Performance testing examined system speed, processing capability, and response time. The efficiency of image processing, CNN prediction time, database retrieval speed, and page loading performance were tested. The aim was to ensure quick disease detection, smooth execution under load, and minimal delays, providing users with a fast and seamless experience.

4. Security Testing :

Security testing ensured the system handled user data safely. Image upload validation, file type restrictions, secure handling of stored data, and prevention of unauthorized access were verified. This testing protected the application from harmful uploads, malicious activities, and data breaches, ensuring the system remains safe, trustworthy, and secure for users.

Conclusion

- The Plant Disease Detection and Supplement Recommendation System is an advanced solution that uses Artificial Intelligence to support modern agriculture. It helps farmers and agricultural professionals detect plant diseases early and receive proper treatment guidance through a web-based platform. The system uses a Convolutional Neural Network (CNN) to analyze leaf images and identify diseases accurately by detecting symptoms like spots, discoloration, and lesions.
- The application follows a modular design with components for image processing, disease detection, and supplement recommendations. It is built using Flask and a secure database, ensuring smooth performance and user-friendly interaction.
- Beyond detection, the system provides detailed information about diseases, including causes, symptoms, prevention methods, and fertilizer suggestions. This helps farmers take timely actions, reduce crop loss, and improve productivity.
- The system also ensures data security through authentication and secure storage. However, its accuracy depends on the quality of training data and it does not consider environmental factors like soil and climate.
- Overall, the system is a reliable and scalable tool that promotes smart farming, sustainable agriculture, and efficient crop management.